

HAMAR II, Norway

WMO: 013860

Lat: 60.801N

Lon: 11.098E

Elev: 132

StdP: 99.75

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-18.0	-22.6	0.5	-20.3	-20.1	0.6	-17.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	26.8	16.3	24.6	15.6	22.7	14.8	17.8	23.6	17.0	22.0	16.1	20.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.4	19.3	14.9	10.7	18.8	14.0	10.2	18.1	50.7	23.7	48.2	22.2	45.5	21.0	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-22.3	28.8	4.2	1.7	-25.3	30.0	-27.8	31.0	-30.2	32.0	-33.2	33.2
			WB	-23.2	19.0	4.3	1.2	-26.3	19.8	-28.8	20.6	-31.3	21.3	-34.4	22.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-6.5	-4.6	-0.1	4.8	10.9	14.7	17.3	15.2	11.3	5.0	0.1	-4.9
	DBStd	9.15	5.40	5.18	4.69	3.51	4.37	2.63	2.94	2.32	2.84	3.53	4.38	6.16
	HDD10.0	2375	512	409	313	159	42	1	0	1	19	159	297	462
	HDD18.3	4785	771	643	572	406	234	113	57	99	211	412	547	720
	CDD10.0	666	0	0	0	4	69	142	226	162	58	5	0	0
	CDD18.3	35	0	0	0	0	3	4	25	3	0	0	0	0
	CDH23.3	322	0	0	0	0	41	42	224	15	0	0	0	0
	CDH26.7	59	0	0	0	0	4	4	51	0	0	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	528	29	23	26	25	37	59	73	68	61	52	43	31
	PrecMax	723	71	72	70	67	76	141	166	146	117	97	77	71
	PrecMin	376	1	2	0	0	7	6	24	10	14	7	12	6
	PrecStd	92	17	14	15	17	20	36	31	39	27	28	20	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	6.6	13.1	19.5	27.1	27.1	30.0	25.0	21.4	16.7	11.3	8.0
		MCWB	2.5	3.4	6.7	10.4	15.0	15.5	17.9	16.6	13.8	13.1	8.8	5.5
	2%	DB	3.5	4.8	10.4	16.6	24.1	24.4	28.2	23.4	19.8	14.0	9.0	6.1
		MCWB	2.1	2.9	5.2	9.0	13.7	14.6	17.3	15.4	13.6	10.3	7.0	4.0
	5%	DB	2.2	3.2	8.3	13.9	21.4	22.1	26.1	21.9	18.2	12.4	7.5	4.5
		MCWB	0.9	1.8	3.9	7.8	12.9	13.5	16.5	14.8	13.1	9.4	5.9	2.5
	10%	DB	1.0	2.3	6.3	11.5	18.7	20.4	23.7	20.4	16.6	10.8	5.8	2.9
		MCWB	-0.1	1.1	2.9	5.8	11.8	12.6	16.1	14.3	12.2	8.1	4.6	1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	4.4	7.2	11.1	16.5	17.4	19.4	18.2	15.9	13.4	9.3	5.9
		MCDB	4.2	5.9	12.0	18.4	22.9	21.6	26.0	21.8	18.5	16.1	11.0	7.6
	2%	WB	2.2	2.9	5.6	9.4	15.2	16.2	18.4	17.1	14.8	11.2	7.5	4.2
		MCDB	3.2	4.3	9.7	15.6	21.9	21.3	25.1	21.0	17.9	13.3	8.9	5.7
	5%	WB	1.1	1.9	4.4	7.9	13.7	15.2	17.6	16.3	13.8	9.8	6.1	2.7
		MCDB	2.0	3.0	7.5	13.0	19.2	20.3	23.9	20.0	17.2	11.8	7.3	4.0
	10%	WB	0.1	1.1	3.2	6.4	12.6	14.1	16.7	15.3	12.8	8.5	4.6	1.6
		MCDB	0.9	2.2	6.2	10.5	17.5	19.0	22.2	18.8	15.9	10.5	5.7	2.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.3	9.2	10.2	9.9	9.5	9.8	9.2	9.0	7.2	5.0	5.9
		MCDBR	6.2	7.6	11.1	14.5	14.1	12.7	13.4	11.8	11.5	8.5	7.3	7.4
		MCWBR	5.3	5.8	6.9	7.7	6.0	5.7	5.1	5.4	6.7	5.9	5.7	6.1
	5% WB	MCDBR	6.1	6.7	10.6	13.5	11.8	10.4	11.1	8.8	9.7	7.2	6.7	6.6
		MCWBR	5.4	5.2	7.0	7.4	5.3	5.5	4.7	4.7	6.0	5.6	5.5	5.6
(40) Clear-Sky Solar Irradiance	taub		0.235	0.281	0.311	0.351	0.355	0.347	0.366	0.358	0.325	0.310	0.254	0.203
	taud		2.176	2.274	2.296	2.332	2.428	2.474	2.437	2.454	2.529	2.485	2.263	2.067
	Ebn at Noon		531	715	813	843	870	883	854	832	804	687	503	398
	Edh at Noon		37	64	87	103	101	98	100	91	71	53	34	25
(44) All-Sky Solar Radiation	RadAvg		0.27	0.92	2.42	3.92	4.75	5.37	4.84	3.78	2.47	1.10	0.35	0.17
	RadStd		0.05	0.17	0.24	0.46	0.48	0.46	0.63	0.34	0.30	0.21	0.10	0.01

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page